

Element G: Construction of Prototype

Project Title: MJ's LAST Stand (Echo Rack)

Day 1: 3/19/2025

Description:

Preparing and converting [Tinkercad file](#) to Onshape and Cura Slicer for 3D-printing:

After we made slight modifications and final touches to his 3D model in TinkerCad, we prepared the model for 3D-printing. We split the many parts of the model apart to allow for more efficient and reasonable 3D-printing (for example, printing the string that attaches the gong to the stand is basically impossible especially at a smaller scale). We originally decided on a scale of 1:5, however, it was simply not possible to print this size with the printer available to us (as it simply wouldn't fit). Instead, we printed the model at a scale of 1:10, saving much material and time.

The following are notable settings used for the 3D-print:

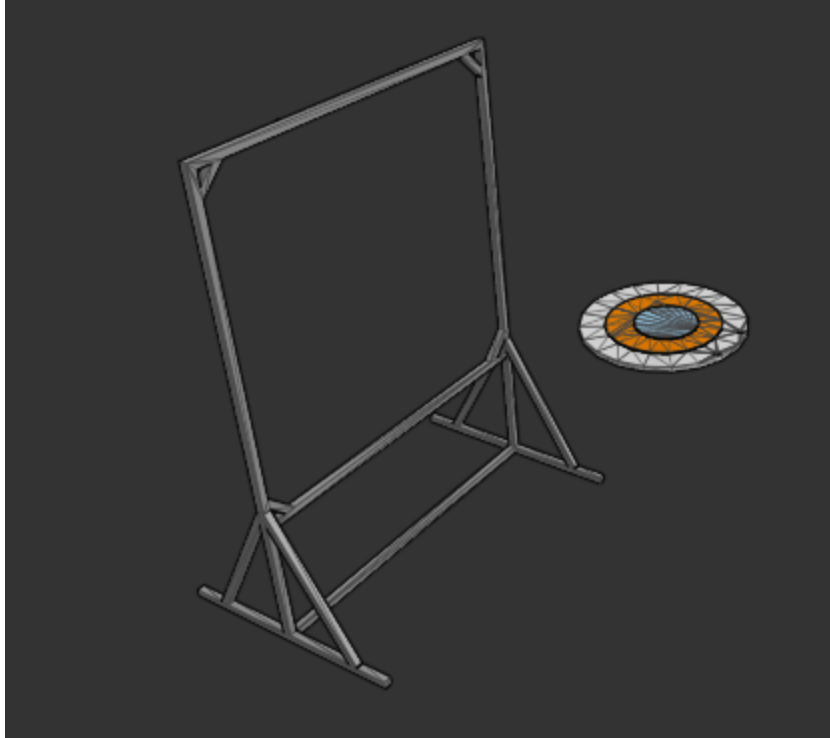
- Speed: 120 mm/s (Standard acceleration set by Cura)
- Temperature: 215°C (Hot end), 55°C (Build plate)
- (Non-tree) Supports Enabled, 15% density, Gyroid pattern
- Brim enabled (especially important for tall prints such as this)
- Layer height: .28mm used with a .4mm nozzle (as the print has no complex geometry, using larger layer heights is definitely reasonable to save time)

The size of the gong was not large enough to use infill.

Print Time: 4 hours 30 minutes

The mute and wheels of the model still need to be figured out and created.

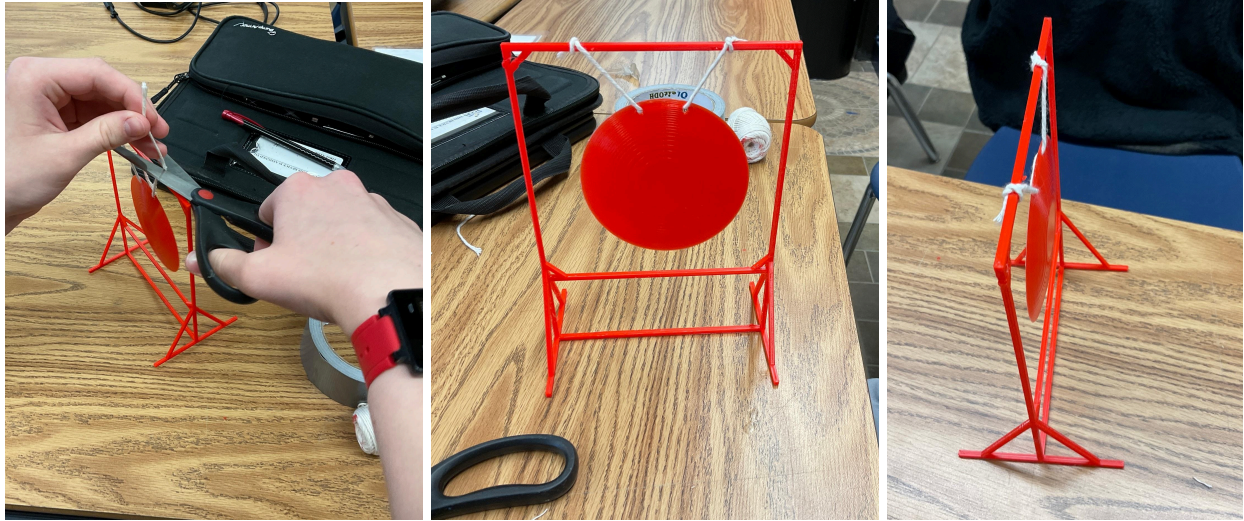
Note: Be really careful taking off supports from the 3D-print, as you do not want to rip or snap part of the stand



March 19th: Screenshot of the CAD model of the Gong Stand and Gong in Onshape, which were printed for the next day

Day 2: 3/20/2025

Description: The 3d printed model came in and we worked on attaching rope and string to the gong stand to help support the gong. We also worked on implementing a mute for the gong stand. As mentioned, today we will work on our mechanical moving parts and how we should implement them in our model. We already know how we will attach it to our final product but the scale model is very small and it will be more difficult to attach our moving parts.

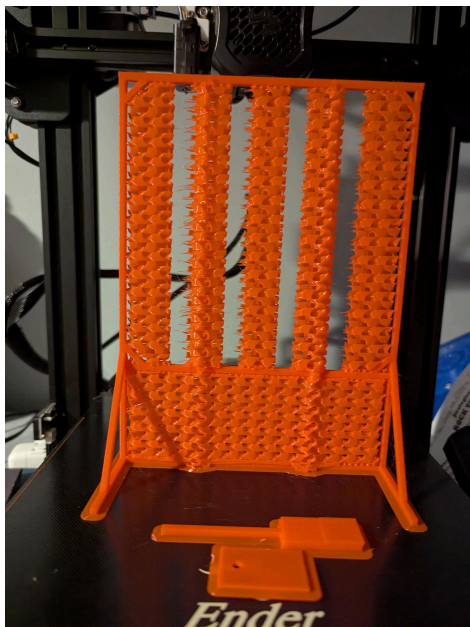


March 20th: Several pictures showing our completed first model/prototype of the gong stand

Later the same day, we spent much time working on the 3D-model, designing the moving parts and mute of the gong stand. The most notable moving part is the hinge on which the mute arms rotate. However, as we have previously noted, the small size of the scale model makes making functional parts much more difficult. In consideration of this, the size of the model has increased from 1:10 to 1:9. Unfortunately, we did not have time to print out all parts of the model (we still need to print the new gong and the hinge itself). The hinge will most likely need to be redesigned larger so it's actually printable for the model.

The only notable change to print settings was the change from 15% support density, to 10% support density.

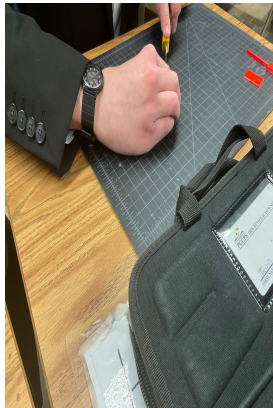
Print time: 5 hours 30 minutes



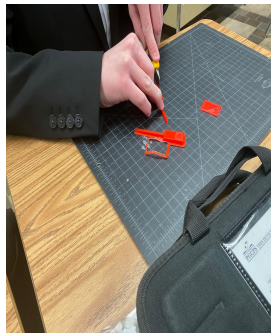
March 21st: Photo of the finished print of the second gong stand prototype, along with new functional parts

Day 3: 3/21/2024

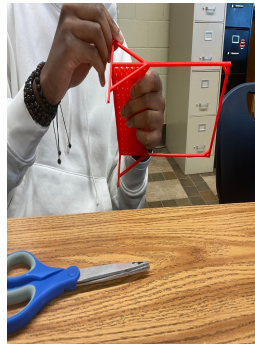
Description: We 3d printed the pedal and the mute for our gong stand. We cut out the 3d models using scissors and a pocket knife. Then we used string, tape and a rubberband to create the mechanism for the mute. After that, we cut out everything and attached all the 3d printed parts to the gong stand.



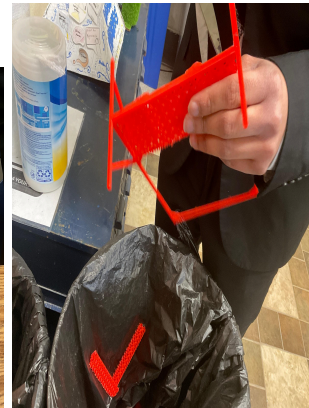
Cutting of excess Filament



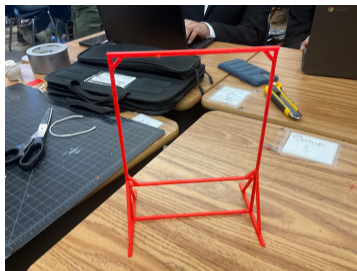
Cutting of excess Filament



Cutting of excess Filament



Cutting of excess Filament

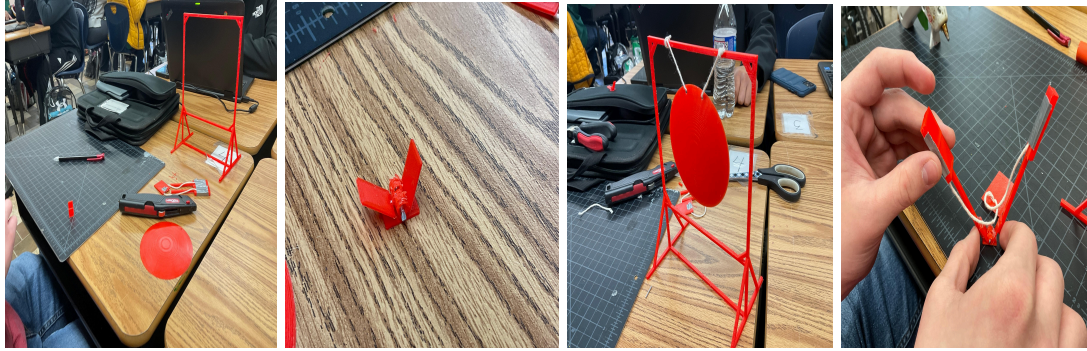


Completed gong stand

Day 4: 3/24/2025

Starting the Mute Prototype

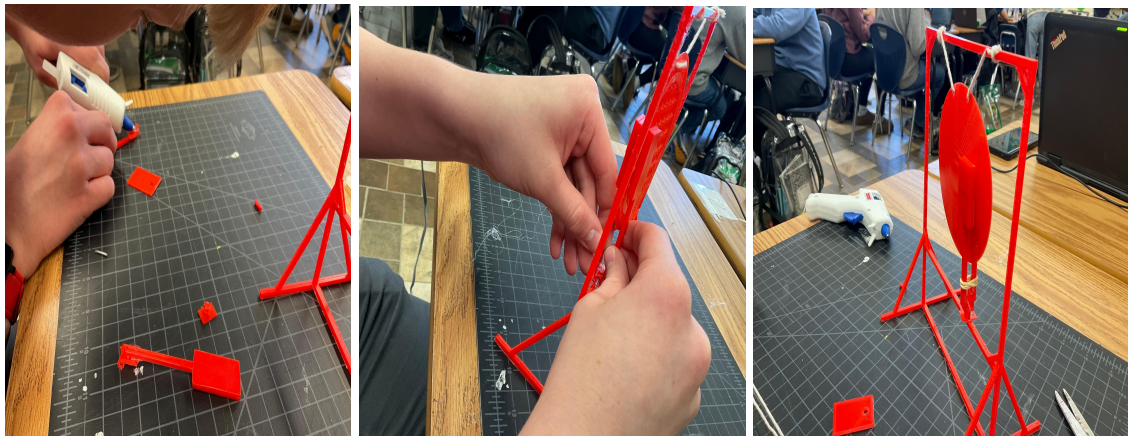
In addition to now having the gong itself (being reprinted to scale once more), we proceeded to assemble the rest of the parts of the prototype including the mute. Making the functional parts of the mute (primarily the hinge) is very difficult for a printer due to its low size. To try mitigating the issues of printing really small geometry, we used as low of a layer height setting as possible: .12mm. However, even shorter should be used (unfortunately, the only nozzle size we had was .4mm). Even after printing at .12mm layer height, the hinge itself was quite weak and did snap. Luckily, we could hot glue it, but it's likely only a temporary fix. Of course, this won't be an issue in the actual gong stand as the hinge will be metal and not underextruded plastic.



March 24th: Several photos showing the various parts of the gong mute being constructed
Description

Day 5: 3/25/2025

Description: We added aa foot pedal to simulate on how on a real gong there would be a foot pedal that might be pressed to also cause more noise/sound to come from a gong during a performance, he used rope/string and popsicle sticks and 3d printed flat bricks to s

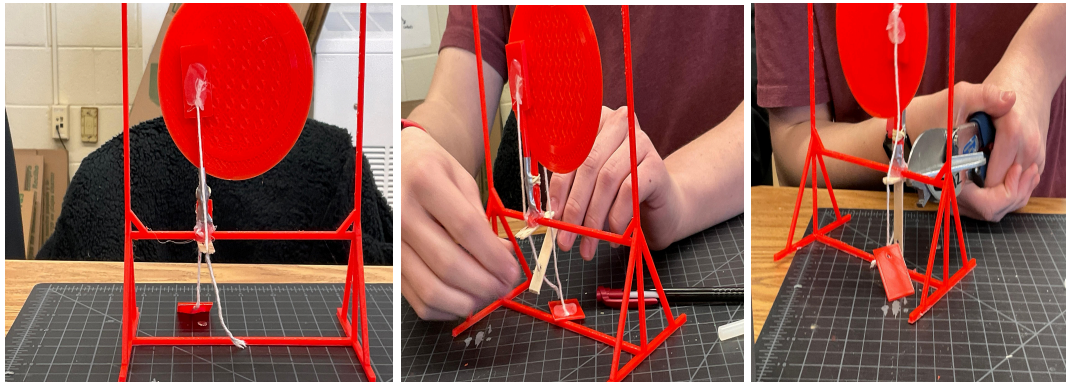


March 25th: These several photos show the build process and progress of the mute prototype throughout the day

Day 6: 3/26/2025

Fine-tuning of the mute prototype

Today, we worked on refining the mute prototype through several ways. We realized when the foot pedal was pressed down, so much force was being used up to force the mute down rather than rotating the mute arms away from the gong. To solve this, we added a piece that would increase leverage and improve the angle at which the strings pull the arms. In addition, we added a vertical support to further prevent force from bending the mute down. With some light testing, we determined that the mute was working well.



March 26th: Photos showing the changes made as described to the mute design